Sebastian Guayacan Mesa

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About Me

Electronics and Electrical Engineer with a minor in Computational Mathematics, experienced in software development for robotics and embedded systems. Proficient in Python and C++ for low-level development, computer vision algorithms, differential robot control, and simulation environments (ROS/Gazebo).

Key experience: • Design and implementation of testing frameworks for embedded systems • Process automation with Python/Bash in Linux environments (Ubuntu) • Continuous integration with GitHub/GitLab CI/CD • Technical documentation of test cases and reports.

I stand out for my analytical skills to address technical challenges, fast learning ability, and adaptability in collaborative environments. I seek to apply these skills in innovative projects integrating robotics, artificial intelligence, and robust system development.

Education

BS **Universidad de los Andes**, Electronics Engineering — Bogotá, Colombia, Jan 2020 – Apr 2025

- Thesis: Dynamic modeling of crops in agrophotovoltaic systems, with distributed generation simulation and stability analysis.
- Minor: Computational Mathematics
- Relevant Courses: Data structures and algorithms, Robotics, Optimization, Learning and evolution of systems.

BS **Universidad de los Andes**, Electrical Engineering — Bogotá, Colombia, Jan 2020 – Apr 2025

- Thesis: Technical-economic analysis of agrophotovoltaic microgrids in non-interconnected zones, including power flow studies and protection sizing.
- Relevant Courses: Elements of electrical systems, Industrial systems, Microgrids

Experience

HERON Agrodrones - Freelance, Systems Developer and Integrator — Bogotá, Colombia, Apr 2025 – Jul 2025

- Design and automation of analysis pipeline for crop visual inspection: Multispectral image processing, specialized index calculation algorithms and models for anomaly detection and health status evaluation.
- Georeferenced data management: Processing architecture for georeferenced images, integration of computer vision tools, optimization and AI frameworks to generate vegetation health maps and recommendations.
- Automation of repetitive processes and standardization of technical documentation to facilitate decision making.

Electrical and Electronics Engineering Department, Universidad de los Andes, Teaching Assistant: Power Systems Economics and Power Electronics — Bogotá, Colombia, Jan 2024 – Dec 2024

- Simulation and analysis of power system operation, including economic dispatch and optimal power flow
- Implementation of projects with real-time simulation (HIL) for power systems and power electronics
- Stability analysis and protection coordination in electrical systems

IBM, Data & AI Portfolio Intern — Bogotá, Colombia, Jan 2024 – Jun 2024

- Collaborated with the digital sales team for Data & AI software products in 9 LATAM countries, providing technical support and operational assistance.
- Performed technical sizing and demonstration of AI solutions, data analysis, process automation (RPA) and virtual assistants.

IBM, Sustainability Portfolio Intern — Bogotá, Colombia — Jul 2023 – Jan 2024

- Provided specialized technical support to the digital sales team for sustainability products in 5 LATAM countries, sizing solutions for enterprise asset management (OMS, ERP, APM, Visual inspection), predictive maintenance with AI and carbon footprint analysis with ESG standards.

Projects

Embedded Systems & Automation — Jun 2022 – Jun 2023

IoT device design and integration (2 combined projects)

- Electronic, software and mechanical integration in automated devices, using Python and C++ for control logic, utilizing various communication protocols.
- Development of Python Interface for remote configuration and monitoring
- Creation of integration manuals and code standardization (Git), focused on replicability and scalability

Robocol — Jun 2021 – Jun 2023

Competitive robotics team leader and STEM outreach

- Development of navigation, computer vision and control algorithms in Python and C++ for motor control and sensors of a differential rover, including electronic circuit design and communication architecture.
- Implementation of navigation algorithms in Python (ROS2) with low-level integration.
- Advanced debugging of concurrency and timing issues in electronic and control systems.

Awards and Honors

CODEFEST AD ASTRA 2023 — May 2023

Second place in Hackathon hosted by Colombian Air Force and Universidad de los Andes.

- Development of video and aerial image analysis algorithms for anomaly detection to protect the Colombian Amazon utilizing Yolo and EasyOCR.
- Development of NLP algorithms for text analysis and classification of public media.

Scholarship: Quiero Estudiar -Universidad de los Andes — Dec 2019

Full scholarship for undergraduate studies in Electronics and Electrical Engineering.

- Awarded for academic excellence and leadership potential.

Skills

Programming Languages: Python, Go, C++, Verilog (basic), MATLAB, SQL, Visual Basic

Tools: Visual Studio Code, Office Suite, Github, ROS2, Gazebo, Simulink, Typhoon HIL - SCADA

Languages: English (Advanced C1), Spanish (Native)

References

Abailable upon request